

Anomaly Detection

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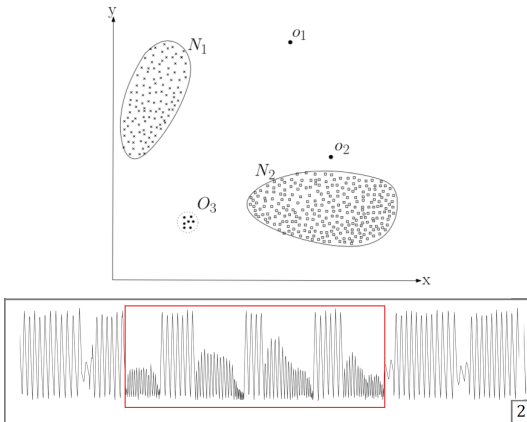
May 2023

Summary

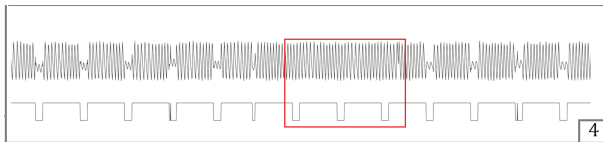
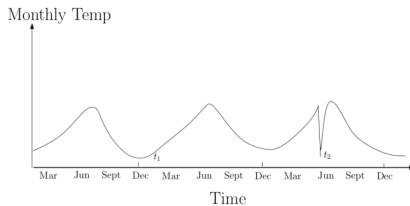
Anomaly detection

- ▶ **Classes of anomalies**
- ▶ **Algorithms**
 - ▶ Distance-based algorithms
 - ▶ LoOF and LOOP
 - ▶ Discords
 - ▶ Domain-based algorithms
 - ▶ One-Class SVM
 - ▶ Isolation Forests
 - ▶ Reconstruction-based algorithms
 - ▶ Subspace-based methods
 - ▶ Neural network-based approaches
 - ▶ Online anomaly detection

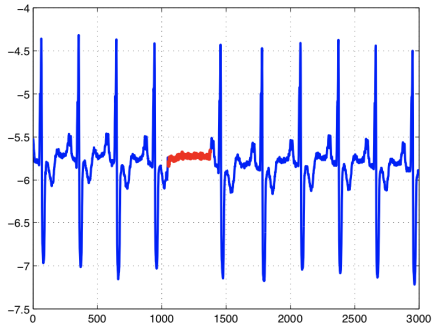
Ponctual Anomalies



Contextual Anomalies



Collective Anomalies



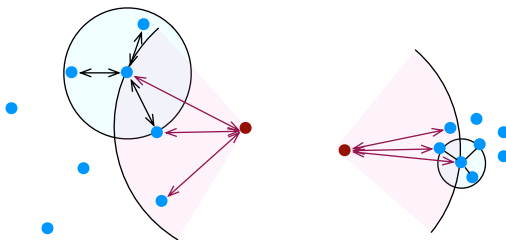
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Local Outlier Factor (LOF) [Breunig, 2000]

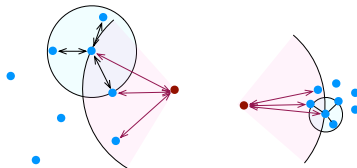
- General principle of k -NN methods: anomalies are far from nominal data and in areas where there are few nominal data



- LOF is based on a “local density” in the neighborhood of each point

$$\mu(\mathbf{x}_i) = \left(\frac{1}{|\mathcal{N}_k(\mathbf{x}_i)|} \sum_{\mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i)} d_k(\mathbf{x}_i, \mathbf{x}_j) \right)^{-1}, \quad \mathcal{N}_k(\mathbf{x}_i): k\text{-NN of } \mathbf{x}_i$$

Local Outlier Factor

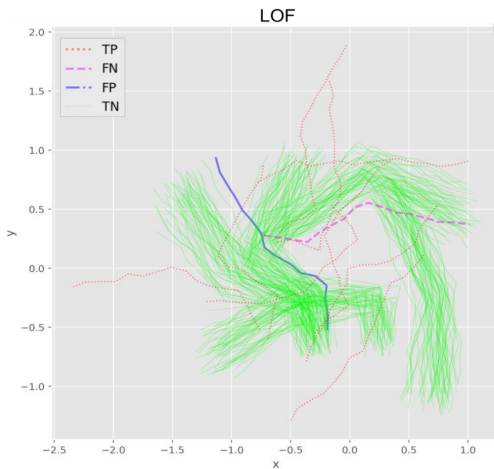


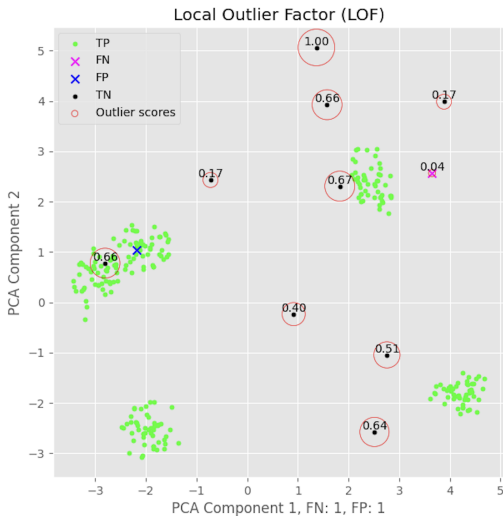
- If the local density of a test point is close to the density of its neighbors, this point is declared as “normal”.

Probabilistic Local Outlier Factor

$$\text{PLOF}_k(\mathbf{x}_i) = \frac{\mu(\mathbf{x}_i)}{\frac{1}{|\mathcal{N}_k(\mathbf{x}_i)|} \sum_{\mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i)} \mu(\mathbf{x}_j)}$$

If $\mathbf{x}_i$ is in a homogeneous area $\text{LOF}_k(\mathbf{x}_i) \approx 1$, else $\text{PLOF}_k(\mathbf{x}_i)$ is “large”.

LOF for Maritime Surveillance ($k = 9$, Contamination = 10/260)

LOF for Maritime Surveillance ($k = 9$, Contamination = 10/260)

Local Outlier Probabilities (LoOP) [Kriegel, 2009]

- ▶ LoOP reformulates LOF in a probabilistic context by **normalizing** $PLOF_k(\mathbf{x}_i)$ and deriving an **anomaly score** $\in]0, 1[$ for each vector $\mathbf{x}_i$:

$LoOP_k(\mathbf{x}_i)$: probability that $\mathbf{x}_i$ is an anomaly

- ▶ **Parameters of LoOP**
 - ▶ **Number of nearest neighbors** k : to be determined by cross validation.
 - ▶ One **significance parameter** λ ensuring that a point o is an outlier for S if

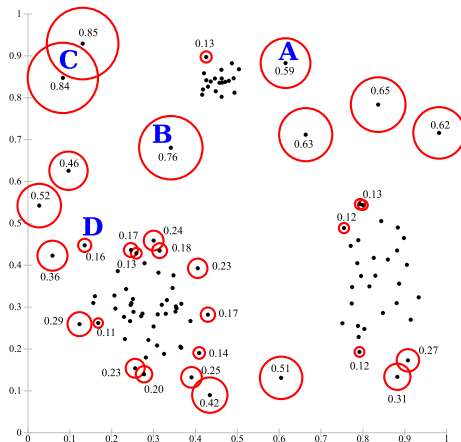
$$P[0 < d(o, s) < \lambda \sigma(o, S)] < \phi, \forall s \in S.$$

where $\sigma(o, S)$ is a kind of average distance between o and the elements of S :

$$\sigma(o, S) = \sqrt{\frac{\sum_{s \in S} d^2(o, s)}{|S|}}.$$

As examples, assuming that $\frac{d(o, s)}{\sigma(o, S)}$ is distributed according to a half $\mathcal{N}(0, 1)$ distribution, we obtain $\lambda = 3$ if $\phi = 99.7\%$ and $\lambda = 2$ if $\phi = 95\%$.

Examples of anomaly probabilities ($k = 20, \lambda = 3$)



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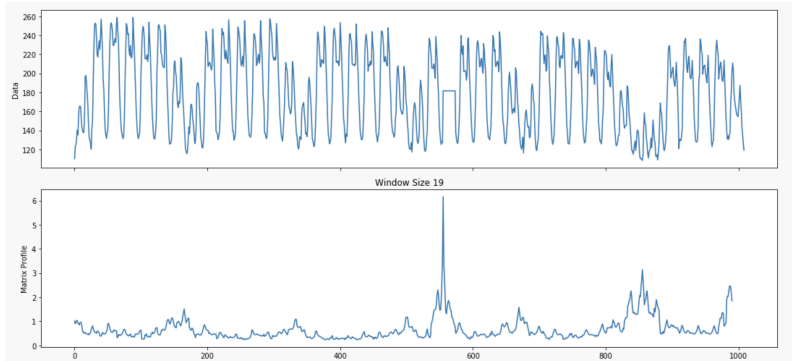
Discords [Keogh, 2005]

- ▶ **Non-Self Match**: M is a non-self match of C at distance of $\text{dist}(M, C)$ if M of length n begins at p , C of length n begins at q and $|p - q| \geq n$.

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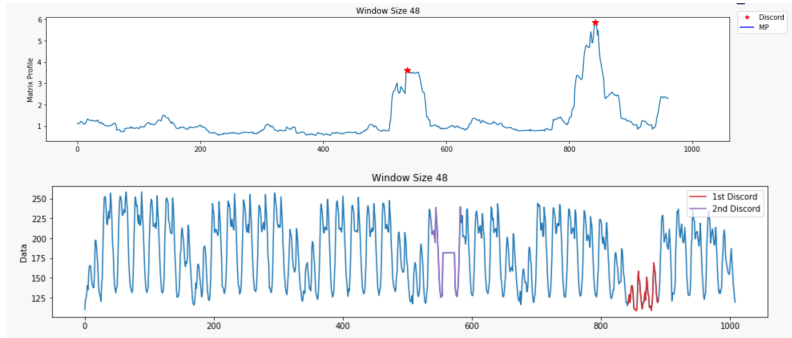
- ▶ **Time Series Discord** : Given a time series T , the subsequence D of length n beginning at position p is called the discord of T , if D has the largest distance to its nearest non-self match.
- ▶ **k th Time Series Discord** : Given a time series T , the subsequence D of length n beginning at position p is called the k th-discord of T if D has the k th largest distance to its nearest non-self match.

One discord



Discord for the hourly power electrical demand in an Italian city during 42 days (1008 hours) - $n = 19$ hours (anomaly size), $k = 1$ (<https://matrixprofile.org/posts/what-are-time-series-discords/>).

Two discords



Discord for the hourly power electrical demand in an Italian city during 42 days (1008 hours) - $n = 48$ hours (anomalies that last 2 days), $k = 2$ (<https://matrixprofile.org/posts/what-are-time-series-discords/>).

Summary

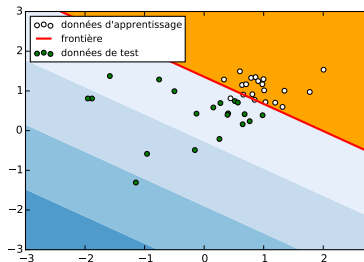
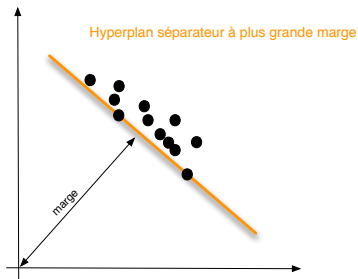
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Linear One-Class-SVM method

- Find the hyperplane separating the training data $\mathcal{X} = \{x_1, \dots, x_n\}$ from the origin and located as far as possible from the origin
- Distance between a point $x = (x, y)^T$ and a straight line $\mathcal{D}$ of equation $\alpha x + \beta y - \rho = 0$

$$d(x, \mathcal{D}) = \frac{|\alpha x + \beta y - \rho|}{\sqrt{\alpha^2 + \beta^2}} = \frac{|w^T x - \rho|}{\|w\|}$$



Linear One-Class-SVM method

- By noting that the margin is $d(\mathbf{0}, \mathcal{D}) = \frac{\rho}{\|\mathbf{w}\|}$, we can solve the following optimization problem ("Soft-margin" SVM classifier)

$$\begin{aligned} &\text{minimize } \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \\ &\text{with the constraints } \mathbf{w}^T \mathbf{x}_i \geq 1 - \xi_i, \xi_i \geq 0, \forall i \end{aligned}$$

or the ν -SVM formulation

$$\begin{aligned} &\text{Minimize } \frac{1}{2} \|\mathbf{w}\|^2 + \frac{1}{n\nu} \sum_{i=1}^n \xi_i - \rho \\ &\text{with the constraints } \mathbf{w}^T \mathbf{x}_i \geq \rho - \xi_i, \xi_i \geq 0, \forall i, \rho \geq 0 \end{aligned}$$

ensuring that the percentage of vectors violating the constraint $\mathbf{w}^T \mathbf{x}_i - \rho \geq 1$ is upper-bounded by ν and that the fraction of support vectors is lower bounded by ν .

Optimization

Lagrangian

$$L(\tilde{\mathbf{w}}, \boldsymbol{\xi}, \boldsymbol{\alpha}, \beta, \rho) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + \frac{1}{n\nu} \sum_{i=1}^n \xi_i - \rho - \sum_{i=1}^n \alpha_i (\mathbf{w}^T \mathbf{x}_i - \rho + \xi_i) - \sum_{i=1}^n \beta_i \xi_i$$

Set to zero the partial derivatives of L with respect to the primal variables $\mathbf{w}$, $\boldsymbol{\xi}$ and ρ to zero yields

$$\mathbf{w} = \sum_{i=1}^n \alpha_i \mathbf{x}_i, \sum_{i=1}^n \alpha_i = 1 \quad \text{and} \quad \alpha_i = \frac{1}{n\nu} - \beta_i \leq \frac{1}{n\nu}, \forall i$$

Kühn and Tucker multipliers

For a convex optimization problem (convex function $f(\mathbf{x})$ to optimize and convex constraints $G_i(\mathbf{x}) \leq 0$), an optimality condition is the existence of parameters $\alpha_i \geq 0$ such that the Lagrangian derivative is zero, i.e.,

$$L'(\mathbf{x}) = f'(\mathbf{x}) + \sum_{i=1}^n \alpha_i G'_i(\mathbf{x}) = 0$$

Dual problem

Solve $L'(x) = 0$

$$w = \sum_{\text{Support vectors}} \alpha_i x_i = x^T \alpha \quad (1)$$

with $\alpha = (\alpha_1, \dots, \alpha_n)^T$, $x = (x_1, \dots, x_n)^T$ and

$$\begin{cases} \alpha_i = 0 & \text{if the constraint is a strict inequality} \\ \alpha_i > 0 & \text{if the constraint is an equality} \end{cases}$$

After replacing the expression of w in the Lagrangian, we obtain

$$U(\alpha) = -\frac{1}{2} \alpha^T (xx^T) \alpha$$

that has to be maximized in the domain defined by $\sum_{i=1}^n \alpha_i = 1$ and $0 \leq \alpha_i \leq \frac{1}{n\nu}$.

Remarks

Simple optimization problem

- ▶ Quadratic (hence convex) function to optimize and linear constraints
- ▶ **Expression of ρ** : the constraints are equalities when $\alpha_i > 0$ and $\beta_i > 0$:

$$\rho = \mathbf{w}^T \mathbf{x}_i = \sum_{j=1}^n \alpha_j \mathbf{x}_j^T \mathbf{x}_i.$$

- ▶ **Classification rule for a vector $\mathbf{x}$**

$$f(\mathbf{x}) = \text{sign} \left(\sum_{\mathbf{x}_i \text{ support vectors}} \alpha_i \mathbf{x}_i^T \mathbf{x} - \rho \right)$$

where the summation is reduced to the support vectors.

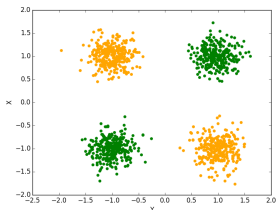
- ▶ ν is a **lower bound for the fraction of support vectors** and an **upper bound for the number of vectors lying outside the separating hyperplane**
- ▶ Generalization to **nonlinear separating curves using kernels** straightforward

Non-linear SVM methods: example 1

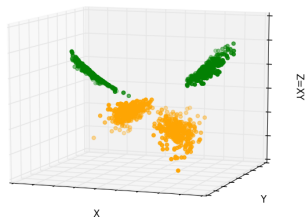
- ▶ Two classes centered around $\{(1, 1)^\top, (-1, -1)^\top\}$ and $\{(1, -1)^\top, (-1, 1)^\top\}$.
- ▶ Training vectors are transformed using the application ϕ

$$\begin{aligned}\phi: \mathbb{R}^2 &\longrightarrow \mathbb{R}^3 \\ \mathbf{x}_i = (x_{i,1}, x_{i,2})^\top &\longmapsto \phi(\mathbf{x}_i) = (x_{i,1}, x_{i,2}, x_{i,1}x_{i,2})^\top\end{aligned}$$

- ▶ A linear separator $\mathbf{w} = (0, 0, 1)^\top$ in the transformed space can separate the data from the two classes



(c) Original data $\mathbf{x}_i$ (Class #1: orange, Classe #2: green).



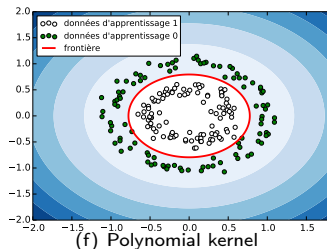
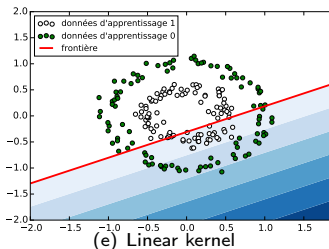
(d) Transformed data $\phi(\cdot)$

Non-linear SVM methods: example 2

- ▶ Two classes defined by two different rings
- ▶ Polynomial transformation ϕ

$$\begin{aligned}\phi : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R}^3 \\ \mathbf{x}_i = (x_{i,1}, x_{i,2})^\top &\longmapsto \phi(\mathbf{x}_i) = (x_{i,1}^2, x_{i,2}^2, \sqrt{2}x_{i,1}x_{i,2})^\top\end{aligned}$$

- ▶ A linear separator $\mathbf{w} = (1, 1, 0)^\top$ in the transformed space corresponds to a “circular” separation in the original space.



Non-linear one-class SVM methods

- ▶ For two data points $\mathbf{x}_i$ and $\mathbf{x}_j$, we have

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \langle \mathbf{x}_i, \mathbf{x}_j \rangle^2.$$

The one-class SVM only needs **scalar products** between the vectors $\mathbf{x}_i$ to be computed!

- ▶ Transposition in the ϕ domain by replacing the scalar product by a **kernel**

$$\langle \mathbf{x}_i, \mathbf{x}_j \rangle \longrightarrow \kappa(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$$

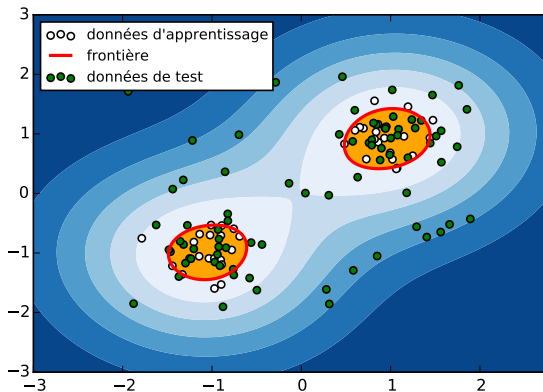
Thus, the transformed vectors $\phi(\mathbf{x}_i)$ and $\phi(\mathbf{x}_j)$ do not need to be computed.

- ▶ **Gaussian kernel**

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp \left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \right).$$

For this example, one can show that the space spanned by $\phi(\mathbf{x})$ has **infinite dimension**.

Non-linear one-class SVM methods



Parameters for the one-class SVM method

Decision rule

$$f(x) = \text{signe} \left(\sum_{i=1}^N \alpha_i \kappa(\mathbf{x}_i, \mathbf{x}) - \rho \right)$$

For the Gaussian kernel

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp \left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2 \right). \quad (2)$$

Effect of the different parameters

- ▶ γ is related with the **regularity of the separating curve**
- ▶ ν allows the **the percentage of outliers from the nominal class** (located outside the separating curve) to be adjusted

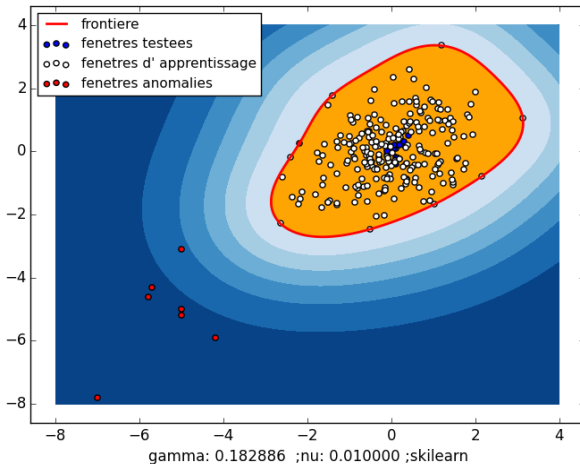
Hyperparameter estimation

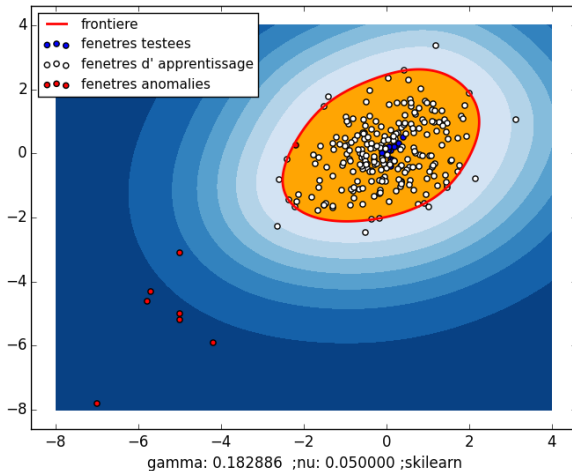
Hyperparameter ν

- ▶ **Expert** or cross validation

Hyperparameter γ

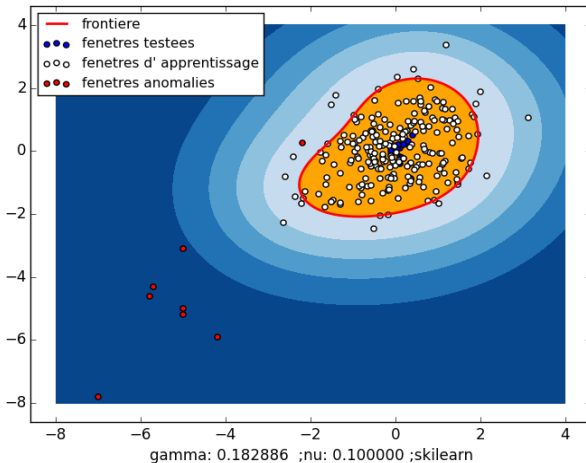
- ▶ Inverse of the number of descriptors (very adhoc)
- ▶ Cross validation
- ▶ “**Trick (Jaakkola, Aggarwal, ...)**”: $\gamma = \frac{1}{2\sigma^2}$ with σ the median of the distances between nominal data
- ▶ More sophisticated methods are available in the literature

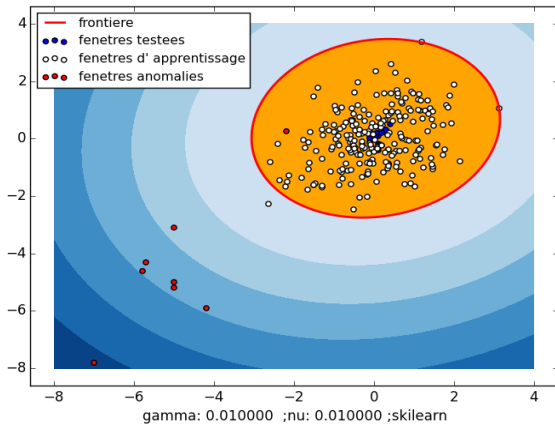
Effect of parameter ν ($\gamma = 0.18$) $\nu = 0.01$ 

Effect of parameter ν ($\gamma = 0.18$) $\nu = 0.05$ 

Effect of parameter ν ($\gamma = 0.18$)

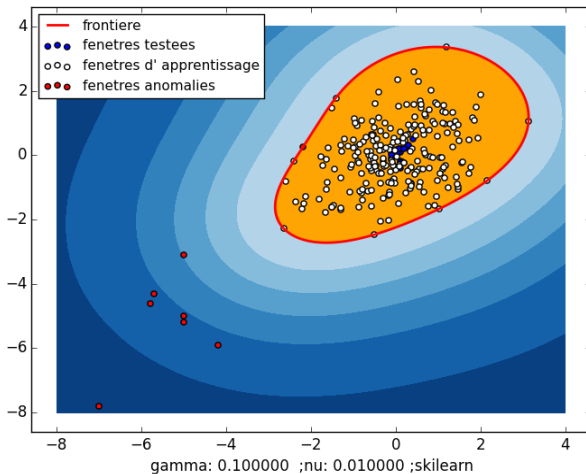
$$\nu = 0.1$$

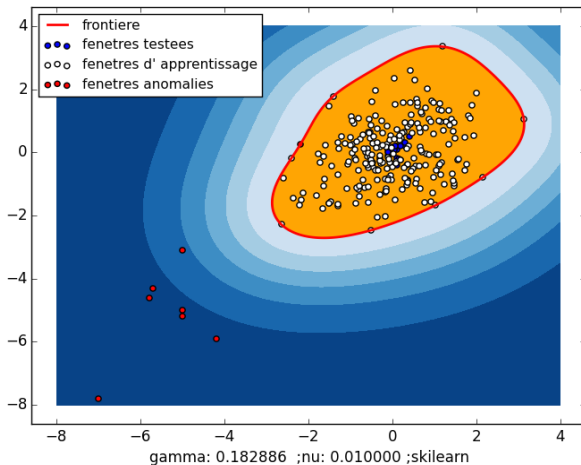


Effect of parameter γ ($\nu = 0.01$) $\gamma = 0.01$ 

Effect of parameter γ ($\nu = 0.01$)

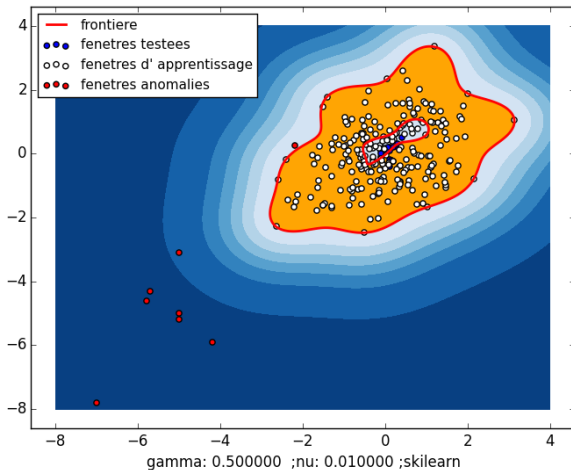
$$\gamma = 0.1$$



Effect of parameter γ ($\nu = 0.01$) $\gamma = 0.18$ (Jaakkola)

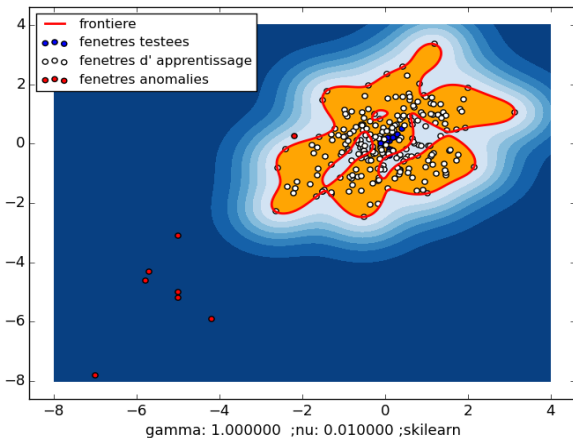
Effect of parameter γ ($\nu = 0.01$)

$$\gamma = 0.5$$



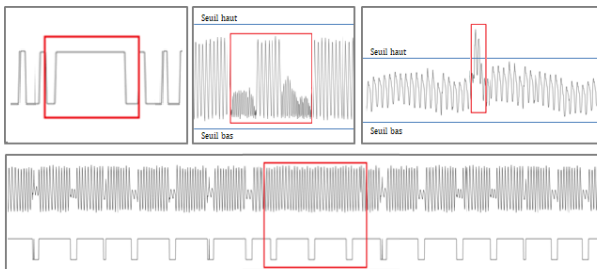
Effect of parameter γ ($\nu = 0.01$)

$$\gamma = 1$$



Application to the analysis of telemetry

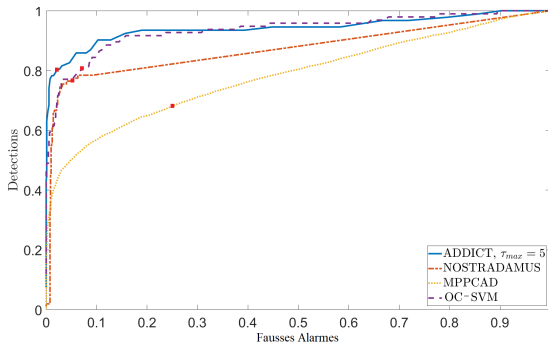
Thesis of B. Pilastre (Nov. 2020)



- ▶ Thousands of telemetry signals
- ▶ Discrete and continuous data
- ▶ Univariate and multivariate anomalies
- ▶ The out of limit (OOL) rule is simple but not efficient!

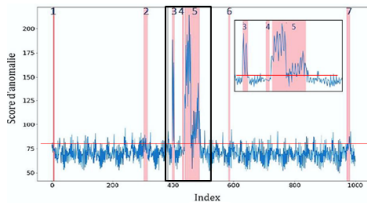
Application to the analysis of telemetry

Receiver operational characteristics

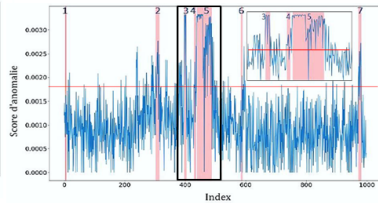


Method	Threshold	P_D	P_{FA}
OC-SVM	0.0018	80.85%	7%
MPPCAD	79.6	80%	13%
NOSTRADAMUS	29	77.26%	6%
ADDICT ($\tau_{max} = 5$)	4.2	80%	3%

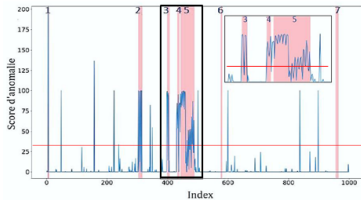
Detected anomalies



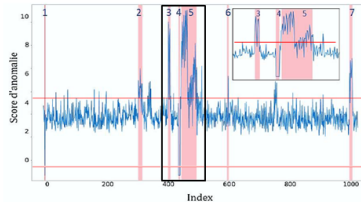
(b) MPPCAD



(c) OC-SVM



(a) NOSTRADAMUS

(d) ADDICT ($\tau = 5$)

Generalization to a semi-supervised scenario

Introduction of a user feedback

- ▶ **Semi-supervised** context: unlabelled data $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and labelled data $\mathcal{Y} = \{\mathbf{y}_1, \dots, \mathbf{y}_m\}$ (e.g., resulting from user feedback)
- ▶ **One-class SVM with user feedback**

$$\text{Minimize } \frac{1}{2} \|\mathbf{w}\|^2 - \rho + C_1 \sum_{i=1}^n \xi_i + C_2 \sum_{j=1}^m \delta_j$$

with the constraints

$$\mathbf{w}^T \mathbf{x}_i \geq \rho - \xi_i, \quad \xi_i \geq 0, \forall i,$$

$$\mathbf{w}^T \mathbf{y}_j \geq \rho - \delta_j, \quad \delta_j \geq 0, \forall j$$

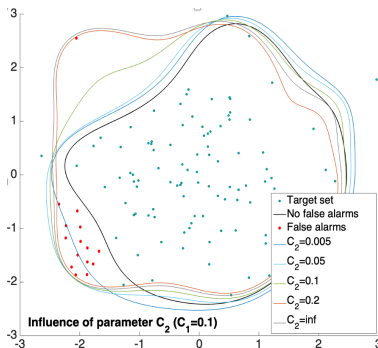
$$\rho \geq 0$$

and C_1, C_2 are two hyperparameters to be adjusted

Hyperparameter selection

Rules

- ▶ $C_1 = \frac{1}{n\nu}$ and $C_2 = 0$, no user feedback
- ▶ $C_1 = C_2$: same importance to unlabelled and labelled data
- ▶ $C_2 \gg C_1$: more importance to user feedback



Support Vector Data Description (Tax and Duin, 1999)

Find a sphere of center $\mathbf{c}$ and radius R that encloses most of the data objects.

Optimization problem

$$\text{minimize } R^2 + C \sum_{i=1}^n \xi_i$$

with the constraints $(\mathbf{x}_i - \mathbf{c})^T (\mathbf{x}_i - \mathbf{c}) \leq R^2 + \xi_i$ $\xi_i \geq 0, \forall i$

Optimization

Lagrangian

$$L(R, \mathbf{c}, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = R^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i \left[R^2 + \xi_i - (\mathbf{x}_i - \mathbf{c})^T (\mathbf{x}_i - \mathbf{c}) \right] - \sum_{i=1}^n \beta_i \xi_i$$

Set to zero the partial derivatives of L with respect to the primal variables $\mathbf{c}$, R and $\boldsymbol{\xi}$ yields

$$\mathbf{c} = \sum_{i=1}^n \alpha_i \mathbf{x}_i, \sum_{i=1}^n \alpha_i = 1 \quad \text{and} \quad \alpha_i = C - \beta_i \leq C, \forall i$$

Dual problem

After replacing the expression of $\mathbf{c}$ in the Lagrangian, we obtain

$$U(\boldsymbol{\alpha}) = \sum_{i=1}^n \alpha_i \mathbf{x}_i^T \mathbf{x}_i - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \mathbf{x}_i^T \mathbf{x}_j$$

that has to be maximized in the domain defined by $\sum_{i=1}^n \alpha_i = 1$ and $0 \leq \alpha_i \leq C$.

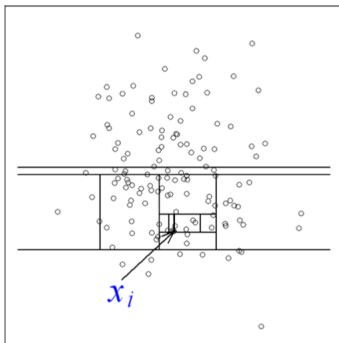
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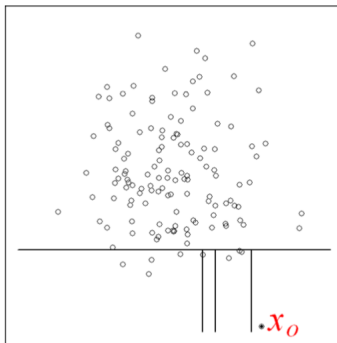
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Principle of isolation forests [Liu, 2008]

- Isolate each point by a random partitioning: **an anomaly will be isolated faster than a nominal point**



(a) Isolating x_i



(b) Isolating x_o

How to built random trees?

Initial strategy proposed in the paper by Liu

For $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ with $\mathbf{x}_i \in \mathbb{R}^d$, a sample of ψ instances $\mathcal{X}' \subset \mathcal{X}$ (ψ : **subsample size**) is used to build an isolation tree.

For each vector $\mathbf{x}_i \in \mathcal{X}'$

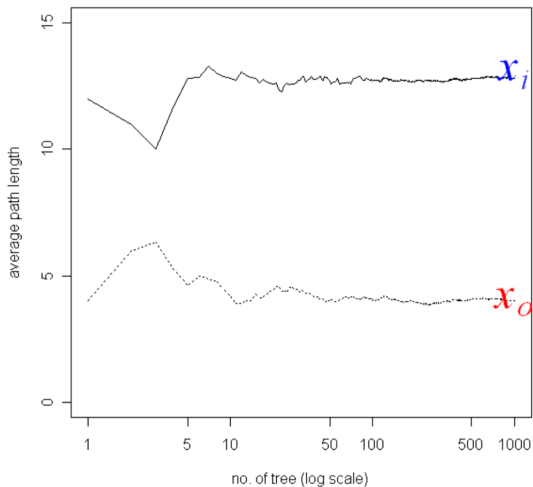
- ▶ Select one feature randomly F_k
- ▶ Compute the minimum and maximum of this feature denoted as $\max_k$ and $\min_k$
- ▶ Divide the space into two parts corresponding to $F_k < \frac{\max_k + \min_k}{2}$ and $F_k > \frac{\max_k + \min_k}{2}$
- ▶ Repeat the process until $\mathbf{x}_i$ has been isolated

Average the numbers of steps obtained with different trees

$$E[h(\mathbf{x}_i)]$$

Note that it is NOT an expectation!!

Length of an average path



(c) Average path lengths converge

Anomaly score

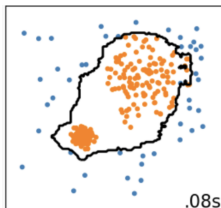
► Definition

$$s(\mathbf{x}_i, \psi) = 2^{-\frac{E[h(\mathbf{x}_i)]}{c(\psi)}}$$

where $E[h(\mathbf{x}_i)]$ is the average path length for $\mathbf{x}_i$ and $c(\psi)$ is the average length of a path for a tree with ψ instances ($c(\psi)$ available in [Liu, 2008])

- if $E[h(\mathbf{x}_i)] = c(\psi)$ then $s(\mathbf{x}_i, \psi) = 0.5$ (uncertainty)
- if $E[h(\mathbf{x}_i)]$ tends to 0, then $s(\mathbf{x}_i, \psi)$ tends to 1 ($\mathbf{x}_i$ is an anomaly)
- if $E[h(\mathbf{x}_i)]$ tends to $\psi - 1$, then $s(\mathbf{x}_i, \psi)$ tends to 0 ($\mathbf{x}_i$ is normal)

► Separating curve: defined using the averaged lengths of the paths



Orange samples: $s(\mathbf{x}_i, \psi) \leq 0.5$, blue samples: $s(\mathbf{x}_i, \psi) > 0.5$.

Summary

Anomaly detection

- ▶ Classes of anomalies
- ▶ Algorithms
 - ▶ Distance-based algorithms
 - ▶ LoOF and LOOP
 - ▶ Discords
 - ▶ Domain-based algorithms
 - ▶ One-Class SVM
 - ▶ Isolation Forest
 - ▶ Reconstruction-based algorithms
 - ▶ Subspace-based methods
 - ▶ Neural network-based approaches
 - ▶ Online anomaly detection

Outlier detection using PCA [Shyu, 2003]

- ▶ Robust estimation of the mean and correlation matrix of normal data
 - ▶ Conventional estimators of the mean and correlation matrix: $\bar{\mathbf{x}}$ and Σ_0
 - ▶ Remove the vectors with the γ th largest values of

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^T \Sigma^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

These vectors are more likely to be **anomalies!**

- ▶ Recompute the mean vector and the correlation matrix Σ of the remaining vectors.
- ▶ Principal component analysis (PCA) of Σ
- ▶ Compute two test statistics from the projected vector $\mathbf{y}_i = \mathbf{P}\mathbf{x}_i$

$$T_{i,q} = \sum_{j=1}^q \frac{y_{ij}^2}{\lambda_j} \quad U_{i,p} = \sum_{j=p-r+1}^p \frac{y_{ij}^2}{\lambda_j}$$

where $\lambda_1, \dots, \lambda_q$ are the q largest singular values of Σ (q such that 50% of the inertia is preserved), and $\lambda_{p-r+1}, \dots, \lambda_p$ are the r smallest values of Σ . Note that $T_{i,q}$ estimates the distance between $\mathbf{x}_i$ and the mean vector whereas $U_{i,p}$ identifies vectors that have correlation structures different from the normal data.

- ▶ Declare that $\mathbf{x}_i$ is an anomaly if $T_{i,q} > c_1$ or if $U_{i,q} > c_2$.

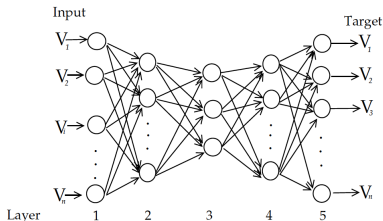
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Outlier detection using RNNs [Hawkins, 2002]

► Architecture of replicator neural networks



- tanh activation functions for layers 2 and 4
- staircase activation function for layer 3
- linear or sigmoidal activation function for the output layer

How to use RNNs for outlier detection?

- **Weights**

The weights of the hidden layers are optimized to minimize the reconstruction error across all training patterns.

$$\frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - o_{ij})^2$$

where m is the number of vectors in the database, n is the number of features of x_i , x_{ij} and o_{ij} are the j th features of the i th data record x_i at the input and output of the network.

- **Outlier factor** for the i th data record

$$\text{OF}_i = \frac{1}{n} \sum_{j=1}^n (x_{ij} - o_{ij})^2.$$

The anomalies are the samples that are not well reconstructed by the network!

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Online anomaly detection

- ▶ **One-class SVM**

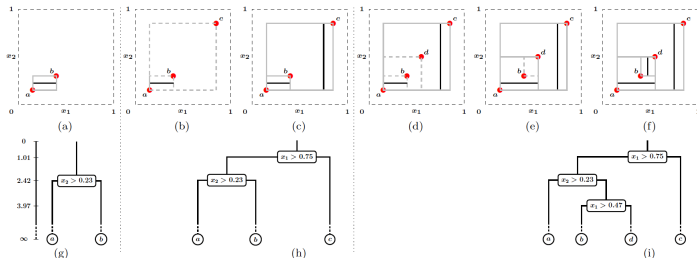
Exploit the structure of the one-class SVM problem to find a subspace minimizer for an $(n + 1)$ -point SVM problem by using the solution of the n -point problem. This can be done using **active-set quadratic programming** (Gao, 2015) or **incremental/decremental learning** (Diehl, 2003)

- ▶ **Online decision trees**

- ▶ **Random Forest (Saffari, 2009)**: Duplicate a new observation (number of replications distributed according to a Poisson $\mathcal{P}(1)$ distribution) and classify these observations using the existing tree. A node is divided into two branches if 1) there is a minimum number of observations in this node, 2) the Gini index is sufficiently reduced after separation. A node is suppressed when its out-of-bag error is too large.

Online anomaly detection

- ▶ One-class SVM
- ▶ Online decision trees
 - ▶ Random Forest (Saffari, 2009)
 - ▶ Mondrian Forests (Lakshminarayanan, 2014): Divide the observation space into hypercubes as a Mondrian painting and update this decision tree when a new observation is arriving by continuing an existing split or by creating new branches inside an existing split.



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Thanks for your attention

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